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PERSPECTIVES
INNOVATIONS

Conducting Systematic Reviews in a Day: Enter Artificial Intelligence

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Systematic reviews (SRs) are the cornerstone of evidence-based decision-making. They synthesize primary research in order to guide practice for clinicians, patients, and policymakers. Yet, despite the central role of SRs, several fundamental challenges limit their timeliness, reproducibility, and overall utility.

Challenges in the Current Landscape

First, SRs are highly time- and cost-intensive. On average, SRs take 67.3 weeks to complete[1] and cost over $140,000 (USD) in total labor[2]. This substantial investment is at odds with the goal of SRs in aiding timely decision-making among patients, clinicians, and policymakers. For example, during the COVID-19 pandemic, early SRs suggesting that ivermectin (an antiparasitic drug) reduced mortality in patients with COVID-19 took ~1 to 2 months to be updated, despite the availability of new practice-changing information 6 months earlier[3,4]. There is an urgent need to provide interest holders with robust evidence in a more timely manner.

Second, although the number of published SRs has grown considerably (an over 20-fold increase in the past 2 decades!)[5], this tremendous growth is not without inefficiencies. Many SRs are redundant, with overlapping questions and objectives and major differences in methodological quality[6]. This can lead to confusion when redundant SRs reach conflicting conclusions, undermining the credibility and utility of evidence synthesis as a whole.

Third, several aspects of today's SR methodologies are not reproducible. In their survey of 100 SRs, Rethlefsen et al. found that only 1% of search strategies could be replicated[7]. Similarly, Hamilton et al.[8] reproduced 10 SRs and found that 8 (80%) had a non-reproducible meta-analysis due to discrepancies in article screening and data extraction. The underlying data are also rarely shared in accessible formats. This lack of transparency prevents error-checking, external validation, reproducibility, and cumulative improvement in the quality of SRs across research groups.

Finally, although evidence accumulation is a dynamic process, the majority of SRs today only offer a static snapshot of evidence (i.e., at 1 point in time). The protracted timelines for both conducting and peer-reviewing an SR mean that, by the time of publication, many SRs are already outdated. In fact, research has shown that 7% of SRs are out of date by the time of publication and 21% are out of date within 12 months[9]. The accelerating pace of biomedical research, coupled with rising rates of article retractions, paper mills, and predatory publications, only further destabilizes the evidence base. Clinicians, policymakers, and patients alike may be relying on evidence summaries that are already obsolete or unreliable.

Automation in SRs

These challenges have made 1 fact increasingly clear: traditional approaches to conducting SRs cannot keep pace with the steep trajectory of publications. As a result, researchers have sought to modernize and speed up aspects of the SR process, including through automation. Early efforts, such as DistillerSR, Rayyan, and Abstrackr, relied on pre-generative artificial intelligence (AI)-based approaches, including classifier-based models and Bidirectional Encoder Representations from Transformers (BERT)-derived natural language processing techniques. While these tools offered incremental improvements in efficiency, they often fell short of clinical-grade performance (e.g., high sensitivity in screening) and required substantial human oversight[10]. Furthermore, these efforts were typically aimed at a single, specific process in the SR workflow, such as the screening of titles and abstracts for potential inclusion. They did not provide end-to-end solutions, limiting the use of these tools for rapidly generating new SRs or keeping SRs up to date.

The advent of generative AI and recent advances in large language models (LLMs) with powerful reasoning and language capabilities have unlocked a new frontier for SR automation. In a recent scoping review of preprints and published literature up to February 2024, Lieberum et al.[11] identified 37 studies that evaluated LLM-based approaches for various SR tasks. The majority of the studies involved "prompting" (i.e., providing an LLM with written instructions or examples that guide how it should perform a specific task) GPT-3.5 or GPT-4 (OpenAI) models to perform search- and screening-related tasks. Furthermore, many of these early LLM evaluations relied on basic "zero-shot" prompting, or providing models with minimal instruction. This is akin to judging the speed and

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efficiency of a race car without shifting gears: it underestimates its true potential.

Emerging evidence indicates that prompt and workflow design are key drivers of AI performance. Nori et al., for instance, reported that OpenAI's MedPrompt framework improved GPT-4's accuracy on MedQA from 78.9% in a zero-shot setting to 90.2% when using structured prompts and workflows[12]. Similarly, Cao et al.[13] demonstrated that optimized prompting strategies substantially improved screening performance over zero-shot baselines.

Several frontier AI laboratories (e.g., OpenAI[14], Anthropic[15], and Google[16]) have also introduced "deep research" features, where base models undergo additional reinforcement learning-based training to apply tool-augmented reasoning, including the ability to query the web and synthesize literature summaries. However, these approaches diverge from established SR methodology: the searches are driven by ad hoc agentic queries rather than structured database searching, and they retrieve only a small subset of potentially relevant studies. Similarly, Elicit (formerly Ought)[17] advertises SR features that enable users to enter a natural-language query and receive an AI-generated evidence summary. However, its workflow relies on semantic searches within the Semantic Scholar database and limits users to screening 500 citations at a time. Taken together, most LLM research tools cannot currently replace methods that comply with Cochrane or PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) guidance.

AI has the potential to address nearly every major limitation in delivering knowledge to interest holders. Processes that once required months of effort can be completed in minutes, freeing researchers to focus on higher-order tasks, such as protocol design, result interpretation, and effective knowledge translation.

otto-SR

We recently introduced otto-SR (Otto Science Institute)[18], a generative AI system for automated screening and data extraction that incorporates advanced prompting strategies and agentic LLM workflows. Data from currently unpublished studies involving benchmarking against dual human reviewers suggest that otto-SR achieved superior performance in both screening (otto-SR: 96.7% sensitivity, 97.9% specificity; human: 81.7% sensitivity, 98.1% specificity) and data extraction (otto-SR: 93.1% accuracy; human: 79.7% accuracy) tasks. Most notably, otto-SR reproduced and updated an entire issue of the Cochrane library (12 SRs) in under 2 days, highlighting the potential for automation to accelerate evidence synthesis and to provide decision-makers with timely information. Across these 12 SRs, otto-SR included nearly twice as many eligible studies as the original Cochrane authors (114 versus 64 studies). When meta-analyses were reproduced, 2 reached newly significant results and 1 became statistically insignificant.

Beyond reducing manual workloads and enhancing accuracy, otto-SR can also generate standardized outputs for research reporting. For example, the system automatically formats a typical "Table 1" for an SR, presenting the basic descriptive information (e.g., 6 studies included only males) and demographic information (e.g., 8 of 24 studies were conducted in Canada) of the included studies. Further clarifying work in this area is necessary, but this preliminary work demonstrates the potential of this tool.

Together, these findings underscore the potential of generative AI systems—when rigorously designed, tested, and validated—to deliver clinical-grade automation in evidence synthesis. However, this potential presupposes that SR teams have formulated a well-defined and methodologically robust protocol that adequately considers relevant clinical and methodological issues. Future work involving AI will need to focus on conducting prospective validations of AI-driven SRs and expanding validation across additional review types and clinical domains.

SRs as Living Knowledge Systems

AI has the potential to address nearly every major limitation in delivering knowledge to interest holders. Processes that once required months of effort can be completed in minutes, freeing researchers to focus on higher-order tasks, such as protocol design, result interpretation, and effective knowledge translation. Transparent, auditable AI workflows can strengthen reproducibility by enabling external scrutiny and fostering cumulative methodological improvements across SR research.

In this envisioned future, SRs no longer function as bottlenecks in evidence-based decision-making, but rather evolve into dynamic, "living" knowledge systems that are transparent, are trustworthy, and accelerate clinical translation[19]. In such a scenario, SRs move beyond serving as static, and possibly outdated, vessels of knowledge. Instead, every SR becomes a "living SR" that houses reliable and up-to-date evidence. This is likely to hold appeal for a broad base of interest holders, ranging from clinicians to patients to health policymakers.

Generative AI represents not only an efficiency gain but also a paradigm shift in how evidence is synthesized, updated, and disseminated. But to realize this promise, the field of SR research must begin to thoughtfully and responsibly integrate

these new approaches to ensure that SRs evolve with the pace of modern science.

Rigorous benchmarking is the first critical step. AI developers must construct sufficiently large, diverse, and well-curated data sets that reflect the full complexities of SR tasks spanning multiple scientific domains (e.g., medicine, economics, and social science) and multiple review types (e.g., scoping, prognostic-factor, and comparative observational). These data sets should underpin model training and ensure that systems can generalize beyond narrow or artificially simplified tasks.

Validation should include both retrospective evaluations using completed SRs and prospective deployments in real-time settings. Clinicians and SR researchers must also assess whether published benchmarks are representative of their use cases and scrutinize how the benchmarks were constructed. For example, reference standards produced by the same developers who built the AI system risk circularity and optimism bias, while AI-generated reference standards can introduce systematic errors that inflate downstream performance.

Ongoing monitoring is a second essential pillar. All AI systems are susceptible to data shift, in which real-world inputs drift away from the distributions observed during training[20]. In SRs, this phenomenon may occur as new trial designs, novel interventions, or regulatory changes emerge. A related challenge is pretraining data contamination: as frontier models expand, there is an increasing risk of published benchmark data sets or entire SRs being included in their corpora, leading to inflated test performance that may not translate to real-world use. This issue is particularly acute in SR research, where many benchmark data sets are public and easily scraped.

These risks demonstrate that benchmarking cannot be a single event. Like the SRs they support, AI tools must be continuously validated, and testing data sets should be regularly expanded to include emerging study designs and underrepresented domains. Random human spot-checking of statistically powered samples is another potential safeguard to detect silent failure modes, to verify correct handling of edge cases, and to guide decisions about when retraining or workflow adjustments are needed.

As AI models assume a greater share of the SR workflow, every decision (e.g., every screening judgment, extracted data point, or risk-of-bias assessment) should also be anchored to clear reasoning outputs. Researchers should continue to retain responsibility for methodological decisions, interpretation, and final approval, even as AI accelerates routine tasks. Equally important is version control, in which each update to a living SR is time-stamped, logged, and archived alongside the exact software versions that were used, allowing corrections to propagate transparently.

As the field transitions, future researchers will need fluency in both the traditional methodology for evidence synthesis and the responsible deployment of AI tools. Training in SR methodology should include sessions on the use and implementation of AI throughout the review process; the testing, validation, and benchmarking of AI outputs; and methods for ensuring quality control in hybrid human-AI workflows.

Until AI systems can consistently demonstrate robust and reliable performance in completing SR processes, human expertise ("human in the loop") will remain a critical component of the review process. But that time horizon is likely shorter than we previously thought. ■

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